

Week 8: React Advanced Concepts

1. React Router

React Router is used for navigation between different pages in a React application without refreshing the browser. It helps create single-page applications with multiple routes. Components like `BrowserRouter`, `Routes`, and `Route` are commonly used. This improves user experience and application structure.

2. Forms Handling in React

Forms handling in React is done using controlled components. Input fields are linked with state variables, making data handling easier. This allows real-time validation and smooth form submission processes. It is essential for login, registration, and user input features.

3. `useRef` Hook, `useContext` Hook

The `useRef` hook is used to directly access DOM elements or persist values without re-rendering. The `useContext` hook is used for sharing data across components without passing props manually. These hooks improve state management and component efficiency.

4. React Project Structure & Reusability

A proper React project structure makes code more maintainable and scalable. Reusable components reduce duplication and improve development speed. Organizing folders like components, pages, hooks, and utilities helps in managing larger projects effectively.

Conclusion

This week focused on advanced React concepts. By learning React Router, form handling, hooks like `useRef` and `useContext`, and project structuring, strong frontend development skills were enhanced.